

Exercise series 2:

ER to Relational Model

Introduction to databases

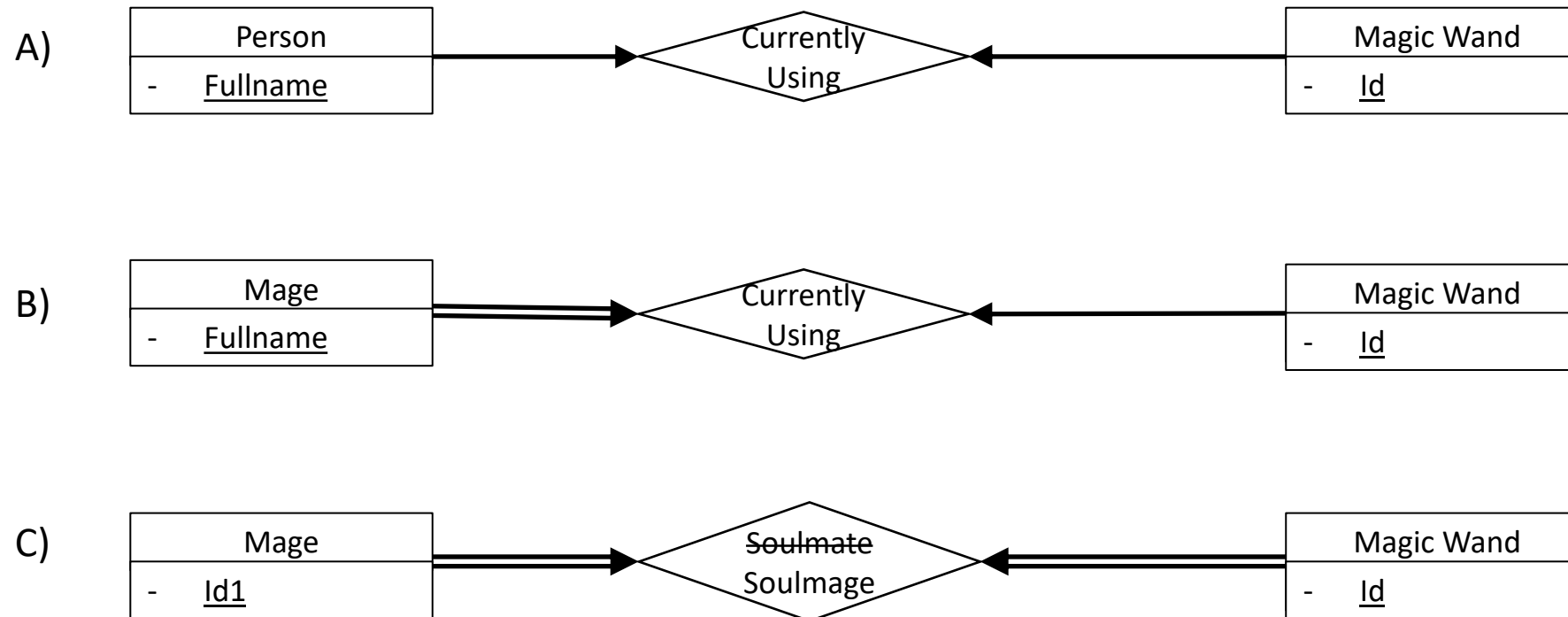


ADReM
Advanced Database Research & Modelling
University of Antwerp

 **Universiteit
Antwerpen**

Problem 1

Using NOT NULL, UNIQUE, FOREIGN KEY and PRIMARY KEY translate these relationships into tables



Problem 2

1. Convert the ER model to the relation model: CREATE TABLE for all entities, CREATE TABLE for many-to-many-relations, and FOREIGN KEY attributes for 1-to-many relations
2. Add example data (entity and many-to-many) to each table using INSERT or UPDATE
3. Retrieve all tuples in table with SELECT: SELECT * FROM tablename;



Problem 2

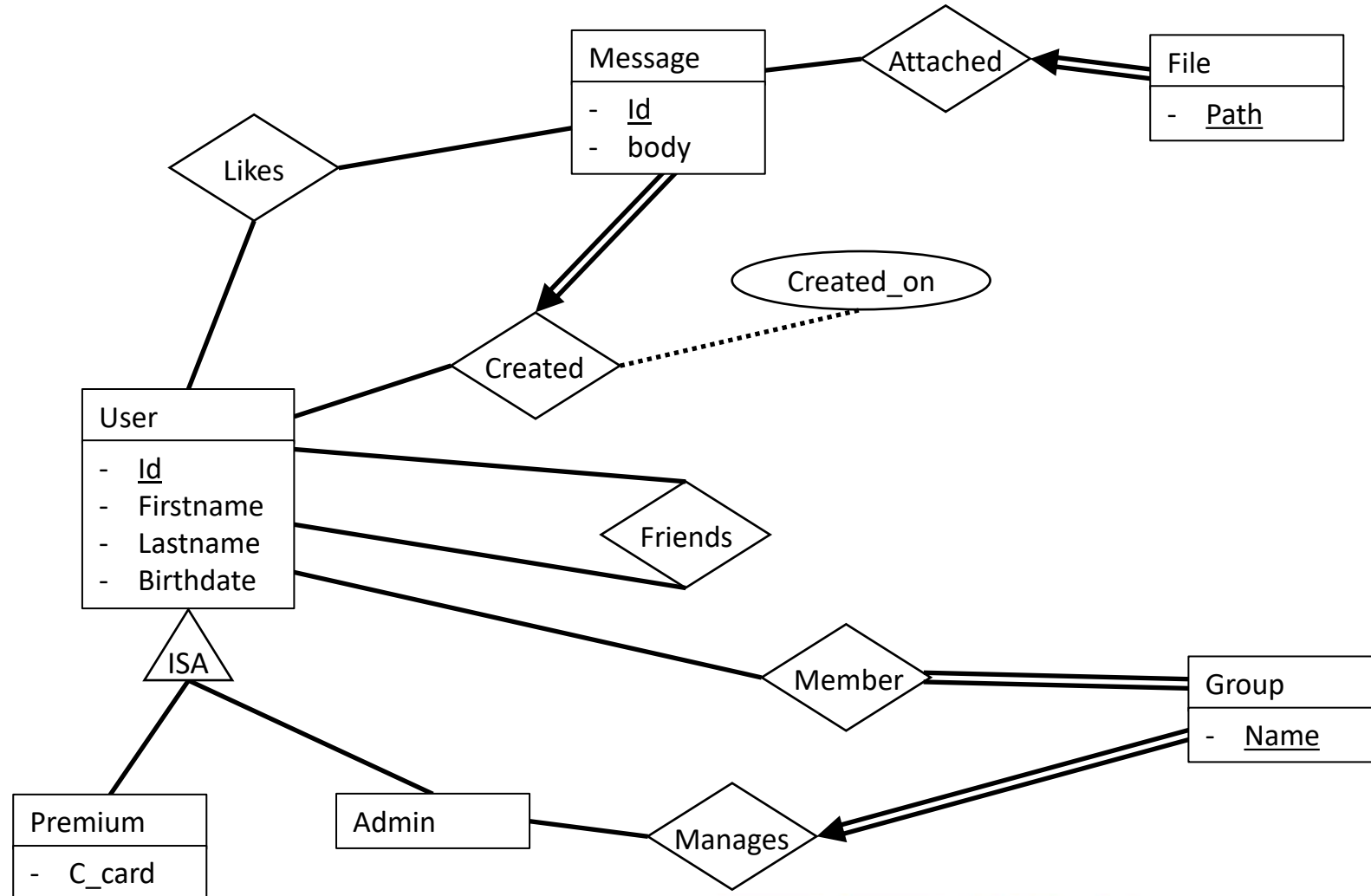
Users are uniquely identified by an email address. For each user we also maintain the first and last name and date of birth. Users can create **messages**, where we keep track of the user that created the message and the date and time. A message has a body (text, possibly using html encoding) and zero or more attached **files**.

Users can also be a member of a **group**. A group has name and is managed by special user who is the group **administrator**. Users can become **friends** with other users. User A asks user B to become friends. After user B accepts the invitation, the friendship status and start date is stored. We also store which friends **like** each message.

Users can also upgrade to a **premium** account with more privileges. For premium users we also store the credit card number.



ER diagram



Problem 3

All the following ER-diagrams have problems when we try to translate them: find those problems

